

Design Q&A and Layer Summary

Extracted from conversation answers on the EmvEnhance refactor.

Q1 - What enhanced the layer model?

PosTerminal is the only public API - it owns card search for every vendor.

EmvBehavior runs EMV only after an entry method is chosen (chip / tap / swipe / manual).

EmvEngine is just an event bus - no business logic.

Vendors live in separate packages behind TerminalFactory; unused host stubs, adapters, and reporters were removed from the wrong layers.

Q2 - Why were CommunicationBehavior and PrinterBehavior inside AbstractEmvBehavior?

They were there for convenience: shared defaults for online authorize and print after EMV steps.

They did not match the job description of EmvBehavior (vendor EMV lifecycle). Host and printer are side services. Correct place: PosTerminal.

Q3 - Do we need AbstractEmvBehavior online/print methods?

No. Those methods were removed:

- onOnlineRequired auto-authorize
- onApproved -> print
- deliverOnlineResult / buildReceipt abstract helpers

AbstractEmvBehavior was deleted. Vendors implement EmvBehavior directly.

Q4 - Where do CommunicationBehavior and PrinterBehavior go?

In abstract PosTerminal:

```
PosTerminal
+-- CommunicationBehavior
+-- PrinterBehavior
+-- searchCard(...)
+-- EmvBehavior (EMV only)
```

Q5 - What is CardSearchListener.Adapter?

A convenience no-op base for CardSearchListener.

The interface has many methods. Adapter implements all as empty so a caller can override only what they care about.

In this project it is unused - PosTerminal uses a full EngineReportingListener. Safe to delete if desired.

Final layer map

Layer	Role
UI	Display; calls `acceptCard` / cancel
PosTerminal	Search, host, printer, orchestration
EmvBehavior	Vendor EMV after entry selected
EmvEngine	Subjects + notify/dispatch
TerminalFactory	Vendor selection only
emvflow	PAX runtime helpers (not core)

Branches / PRs produced

Branch	Focus	PR
`cursor/pax-emv-behavior-bridge-a435`	PaxEmvBehavior bridge, searchCard listener	#2
`cursor/full-postterminal-refactor-a435`	Full PosTerminal-centric refactor	#3
`cursor/package-layout-refactor-a435`	Packages + host on PosTerminal	#4
`cursor/transaction-step-dispatch-4c01`	Earlier step-dispatcher (Branch N baseline)	base